

$$1) a) \log n! < n^{1.54} < n^{\log n}$$

$$\left[\log n! = \log (1+2+\dots+n) \right] \leq \left[\log (n+n+\dots+n) = n \log n \right]$$

$$\log n! = o(n \log n)$$

Comparison 1

$$\lim_{n \rightarrow \infty} \frac{n^{\log n}}{n^{1.54}} = \frac{\log n * n^{(\log n - 1)}}{1.54 * n^{0.54}} = \infty \text{ (for any } n \text{ value)}$$

$$n^{1.54} = o(n^{\log n})$$

Comparison 2

$$\lim_{n \rightarrow \infty} \frac{n \log n}{n^{1.54}} = \frac{n \log n}{n \cdot n^{0.54}} = \frac{\log n}{n^{0.54}} = \frac{(1/\ln n)}{0.54 * n^{-0.46}} = \frac{1}{0.54 * n^{-0.46} * \ln n} = 0$$

$$n \log n = o(n^{1.54})$$

$$b) n^{2/3} < 2^{\log n} < 4^{\log n}$$

Comparison 1

$$\lim_{n \rightarrow \infty} \frac{2^{\log n}}{4^{\log n}} = \frac{2^{\log n}}{(2^{\log n})^2} = \frac{1}{2^{\log n}} = \frac{1}{n} = 0 \quad 2^{\log n} = o(4^{\log n})$$

Comparison 2

$$\lim_{n \rightarrow \infty} \frac{2^{\log n}}{n^{2/3}} = \frac{\log n * 2^{\log n - 1}}{\frac{2}{3} * n^{-1/3}} = \infty \quad n^{2/3} = o(2^{\log n})$$

Hit:

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \rightarrow f(n) = o(g(n))$$

$$= \text{constant} \rightarrow f(n) = \Theta(g(n))$$

$$= \infty \rightarrow g(n) = \Omega(f(n))$$

$$c) \sqrt{n} < 8^{\log n} < 7^{n/2}$$

$$\lim_{n \rightarrow \infty} \frac{8^{\log n}}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{n^{\log 8}}{n^{1/2}} \quad \text{since } a^{\log b^c} = c^{\log b^a}$$

$$= \lim_{n \rightarrow \infty} n^{\log 8 - 1/2}$$

$$= \infty$$

$$\sqrt{n} < 8^{\log n} < 7^{n/2}$$

$$d) 2^n + n^3 < 3^n + n^3 < n^n + 1000n$$

Comparison 1

$$\lim_{n \rightarrow \infty} \frac{2^n + n^3}{3^n + n^3} = \frac{\infty}{\infty} \xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln 2 + 3 \cdot n^2}{3^n \cdot \ln 3 + 3 \cdot n^2} = \frac{\infty}{\infty}$$

$$\xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln^2 2 + 6n}{3^n \cdot \ln^2 3 + 6n} = \frac{\infty}{\infty} \xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln^3 2 + 6}{3^n \cdot \ln^3 3 + 6} = \frac{\infty}{\infty}$$

$$\xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{2^n \cdot \ln^4 2}{3^n \cdot \ln^4 3} = \frac{\ln^4 2}{\ln^4 3} \cdot \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = \frac{\ln^4 2}{\ln^4 3} \cdot 0 = 0$$

Comparison 2

$$\lim_{n \rightarrow \infty} \frac{3^n + n^3}{n^n + 1000n} = \frac{\infty}{\infty} \xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{3^n \cdot \ln 3 + 3 \cdot n^2}{n^n \cdot (\ln(n) + 1) + 1000} = \frac{\infty}{\infty}$$

$$\xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{3^n \cdot \ln^2 3 + 6n}{n^n \cdot (\ln(n) + 1)^2 + n^{n-1}} = \frac{\infty}{\infty}$$

$$\xrightarrow{\text{L'Hospital}} \lim_{n \rightarrow \infty} \frac{3^n \cdot \ln^3 3 + 6}{n^n \cdot (\ln(n) + 1)^3 + 3 \cdot n^{n-1} \cdot \ln(n) + 3 \cdot n^{n-1} - n^{n-2}}$$

$$= \ln^3 3 \cdot \lim_{n \rightarrow \infty} \left(\frac{3}{n}\right)^n \cdot \lim_{n \rightarrow \infty} \frac{1}{(\ln(n) + 1)^3 + 3 \cdot n \cdot \ln(n) + 3 \cdot n - n^{-2}} = \ln^3 3 \cdot 0 \cdot 0 = 0$$

2) Calculate the running time of codes.

a) ($k=1, k \leq J, k = k*3$) run $\rightarrow \log_3 J$

$$\sum_{i=1}^{2n} \sum_{j=1}^i \log_3 J \quad \text{times in total}$$

$$\underbrace{\sum_{i=1}^{2n} \log_3 (i)!}_{\log_3 (i)!}$$

$$\sum_{i=1}^{2n} \log_3 (i)! \leq \sum_{i=1}^{2n} i * \log_3 i \leq \sum_{i=1}^{2n} 2n * \log_3 2n$$

$$2n (2n * \log_3 2n) = 4n^2 \log_3 2n = O(n^2 \log n)$$

b) $k=5 \rightarrow \text{step } 0$

$k=x^5 \rightarrow \text{step } 1$

$k=(x)^{x^5} \rightarrow \text{step } 2$

$k=(x^x)^{x^5} \rightarrow \text{step } 3$

$\vdots$

$k=(x^x)^{a+5} \rightarrow \text{step } a+1$

• How many times the loops run?
We need to find # of steps

$$n = x^{x^{a+5}} \rightarrow a = \underbrace{(\log_x \log_x n) - 5}_{\# \text{ steps}}$$

• Running time of each steps depends on the previous value of k

$$5 + x^5 + x^{x^5} + x^{x^{x^5}} + \dots + x^{x^{x^{a+5}}} \leq \underbrace{x^{x^{a+5}} + x^{x^{a+5}} + \dots + x^{x^{a+5}}}_{n \log \log n}$$

Time complexity is $O(n \log \log n)$

3) a) $3^n \neq o(2^n)$

• $3^n \in o(2^n)$ we need to prove $3^n \leq c \cdot 2^n \quad n \geq n_0 \quad c > 0$

$$\frac{3^n}{2^n} \leq \frac{c \cdot 2^n}{2^n} \quad \left(\frac{3}{2}\right)^n \leq c$$

• when $n \rightarrow \infty$, $\left(\frac{3}{2}\right)^n \rightarrow \infty$. So $\left(\frac{3}{2}\right)^n \leq c$ is not true for all $n \geq n_0$ any c

b) $n^2 + n \neq o(n^2)$

• $T(n) = n^2 + n$, $T(n) = o(n^2) \wedge T(n) \neq \Theta(n^2) \rightarrow T(n) = o(n^2)$

solution 1 $\left\{ \lim_{n \rightarrow \infty} \frac{n^2 + n}{n^2} = \lim_{n \rightarrow \infty} 1 + \frac{1}{n} = 1 + \frac{1}{\infty} = 1 \rightarrow n^2 + n = \Theta(n^2) \right.$
for this, $o(n^2)$ is not true.

solution 2 $\left\{ \begin{array}{l} \cdot \text{check } n^2 + n \in o(n^2) \text{ is true/false} \\ \cdot \text{check } n^2 + n \in \Omega(n^2) \text{ is true/false} \\ \cdot \text{check (little o) rules is suitable } [\in o(n^2) \wedge \notin \Theta(n^2)] \end{array} \right.$

c) $4n^3 - 3n + 1 \neq \Theta(n^2)$

• $\Theta(n^2) = o(n^2) \wedge \Omega(n^2)$

• $4n^3 - 3n + 1 \in o(n^2)$ we need to prove $4n^3 - 3n + 1 \leq c n^2 \quad n \geq n_0, c > 0$

$$n^2 \left(4n - \frac{3}{n} + \frac{1}{n^2} \right) \leq c n^2 \rightarrow \left(4n - \frac{3}{n} + \frac{1}{n^2} \right) \leq c$$

• when $n \rightarrow \infty$, $\underbrace{4n}_{\rightarrow \infty} - \underbrace{\frac{3}{n}}_{\rightarrow 0} + \underbrace{\frac{1}{n^2}}_{\rightarrow 0} \rightarrow \infty$, so c not found.

It's contradiction for our assumption.

4) Master Theorem

$$a) T(n) = 4T(n/2) + n$$

$$\left. \begin{aligned} [f(n)=n] &= O(n^{\log_2 4 - \frac{1}{\epsilon}}) \\ T(n) &= \Theta(n^{\log_2 4}) = \Theta(n^2) \end{aligned} \right\} \text{case 1}$$

$$b) 4T(n/2) + n^2 = T(n)$$

$$\left. \begin{aligned} (f(n)=n^2) &= \Theta(n^{\log_2 4}) \\ T(n) &= \Theta(n^2 \log n) \end{aligned} \right\} \text{case 2}$$

$$c) T(n) = 4T(n/2) + n^2 \log n$$

Can not be solved using Master Theorem.

$$f(n) = n^2 \log n \neq \Omega(n^{\log_2 4 + \epsilon})$$

this $f(n) > n^2$ for this reason lower bound may be $\Omega(n^2)$ (case 3 not suitable)

this $f(n) < n^3$ for this reason upper bound may be $O(n^3)$ (case 1 not suitable)

$$\lim_{n \rightarrow \infty} \frac{n^2 \log n}{n^{2+\epsilon}} = \frac{2n \log n + n}{(2+\epsilon)n^{1+\epsilon}} = \frac{n^1(1+2\log n)}{n(n^\epsilon(2+\epsilon))} = \frac{1+2\log n}{n^\epsilon(2+\epsilon)} = 0 \rightarrow n^1 \log n = O(n^{2+\epsilon})$$

$$d) T(n) = 4T(n/2) + n^3$$

$$\left. \begin{aligned} [f(n)=n^3] &= \Omega(n^{\log_2 4 + \frac{1}{\epsilon}}) \\ T(n) &= \Theta(f(n)) = \Theta(n^3) \end{aligned} \right\} \text{case 3}$$

$$T(n) = a[T(n/b)] + f(n)$$

$$\text{case 1 } f(n) = O(n^{\log_b a - \epsilon})$$

$$T(n) = \Theta(n^{\log_b a})$$

$$\text{case 2 } f(n) = \Theta(n^{\log_b a})$$

$$T(n) = \Theta(n^{\log_b a} \log n)$$

$$\text{case 3 } f(n) = \Omega(n^{\log_b a + \epsilon}), \epsilon > 0$$

$$a f(n/2) \leq c f(n) \quad c > 1$$

$$T(n) = \Theta(f(n))$$

5) Recursive Relations solve with substitution method

$$a) T(n) = 2T(n/2) + n/\log n \quad T(1) = 1$$

$$T(n) = \frac{n}{\log n} + 2 \frac{n/2}{\log n/2} + 4 \frac{n/4}{\log n/4} + \dots + n \frac{n/n}{\log n/n}$$

$$T(n) = \frac{n}{\log n} + \frac{n}{\log \frac{n}{2}} + \frac{n}{\log \frac{n}{4}} + \dots + \frac{n}{1}$$

$$T(n) = n \left[\underbrace{\frac{1}{\log n}}_{\log n - 1} + \underbrace{\frac{1}{\log \frac{n}{2}}}_{\log n - 2} + \frac{1}{\log \frac{n}{4}} + \dots + 1 \right] = n \sum_{i=0}^{\log n - 1} \frac{1}{\log n - i}$$

Let prove by induction. we assume that $n = 2^k$, $k \in \mathbb{N}$ it is true or not.

$$\text{for } k=0, n=2^k=2^0=1 \quad T(n)=T(1)=1 \cdot (1 + \underbrace{\sum_{i=0}^{-1} \frac{1}{r-i}}_{\emptyset}) = 1$$

$$\text{for } k=r, n=2^r \quad T(2^r) = 2^r \cdot \left(1 + \sum_{i=0}^{r-1} \frac{1}{r-i} \right) \quad (\text{hypothesis})$$

$$\text{for } k=r+1, n=2^{r+1} = 2^r \cdot 2$$

$$\begin{aligned} T(n=2^{r+1}) &= 2T(n/2) + n/\log n \\ &= 2T\left(\frac{2^{r+1}}{2}\right) + \frac{2^{r+1}}{\log 2^{r+1}} \\ &= 2 \left[2^r \left(1 + \sum_{i=0}^{r-1} \frac{1}{r-i} \right) \right] + \frac{2^{r+1}}{r+1} \\ &= 2^{r+1} \left(1 + \sum_{i=0}^r \frac{1}{r-i} \right) \end{aligned}$$

$$T(n) = n \sum_{i=0}^{\log n - 1} \frac{1}{\log n - i} = \Theta(n \log \log n)$$

$$b) T(n) = 2T(\sqrt{n}) + 1, T(1) = 1$$

• Introduce a change of variable $n = 2^m$

$$T(2^m) = 2T(\sqrt{2^m}) + 1 = 2T(2^{m/2}) + 1$$

• let us make another change $S(m) = T(2^m)$

$$S(m) = 2S(m/2) + 1$$

$$S(m) = 2(2S(m/4) + 1) + 1 = 2^2 S(m/2^2) + 2 + 1$$

$$\vdots$$

$$S(m) = 2^k S\left(\frac{m}{2^k}\right) + 2^{k-1} + 2^{k-2} + \dots + 1$$

• we assume that $m = 2^k$

$$S(m) = S(2^k) = 2^k S(1) + 2^{k-1} + 2^{k-2} + \dots + 1$$

$$S(1) = T(2^1) = T(2) = 2T(\sqrt{2}) + 1 = 3$$

$$S(m) = 2^k \cdot 3 + \underbrace{2^{k-1} + 2^{k-2} + \dots + 1}_{\sum_{i=1}^k 2^{k-i} = \frac{1-2^k}{1-2} = 2^k - 1}$$

$$S(m) = 3 \cdot \frac{2^k}{m} + \frac{2^k}{m} - 1 = 4m - 1$$

Proof by induction for $S(m) = 4m - 1$

we assume that $m = 2^k$ for all $k \in \mathbb{N}$

$$\text{for } k=0, m=2^0=2^0=1 \quad S(1) = 4-1=3$$

$$\text{for } k=r, m=2^r \quad S(2^r) = 4 \cdot 2^r - 1 = 2^{r+2} - 1 \text{ (hypothesis)}$$

$$\text{for } k=r+1, m=2^{r+1} \quad S(2^{r+1}) = 4 \cdot 2^{r+1} - 1 \text{ (need to prove)}$$

$$S(2^{r+1}) = 2 \cdot S\left(\frac{2^{r+1}}{2}\right) + 1 = 2S(2^r) + 1 = 2[2^{r+2} - 1] + 1$$

$$= 2^{r+3} - 2 + 1 = 2^{r+3} - 1 \quad \checkmark$$

$$S(m) = T(2^m = n) = 4m - 1 = 4 \log n - 1 = \Theta(\log n)$$

$$c) T(n) = T(n/2) + 1, T(1) = 1$$

$$T(n) = T(n/2) + 1$$

$$T(n) = (T(n/4) + 1) + 1$$

$$T(n) = ((T(n/8) + 1) + 1) + 1$$

⋮

$$T(n) = \underbrace{T(n/n)}_1 + \underbrace{1+1+\dots+1}_{\text{how many?}}$$

$$T(n) = T\left(\frac{n}{2^k}\right) + \underbrace{1+1+\dots+1}_{k \text{ terms}}$$

we assume that $n = 2^k, k = \log n$

$$T(n) = 1 + k = \log n + 1$$

let prove by induction

$$\text{for } k=0, n=2^0=2^0=1 \quad T(n) = T(1) = \log 1 + 1 = 1 \quad \checkmark$$

$$\text{for } k=r, n=2^k=2^r \quad T(n) = T(2^r) = \log 2^r + 1 = r + 1 \quad (\text{hypothesis})$$

$$\text{for } k=r+1, n=2^{r+1} \quad T(n) = T(2^{r+1}) = \log 2^{r+1} + 1 \quad (\text{need to prove})$$

$$T(2^{r+1}) = T\left(\frac{2^{r+1}}{2}\right) + 1$$

$$= T(2^r) + 1$$

$$= r + 1 + 1 = r + 2 \quad \checkmark$$

$$T(n) = \log n + 1 = \Theta(\log n)$$

$$d) T(n) = 8T(n/2) + n^2 \quad T(1) = 1$$

$$\cdot T(n) = 8 \left[8T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)^2 \right] + n^2 = 8^2 T\left(\frac{n}{2^2}\right) + 2n^2 + n^2$$

$$T(n) = 8^3 T\left(\frac{n}{2^3}\right) + 2^2 n^2 + 2n^2 + n^2$$

⋮

$$T(n) = 8^k T\left(\frac{n}{2^k}\right) + n^2 (2^k - 1)$$

• We assume that $n = 2^k$, $k = \log n$

$$T(n) = n^3 \cdot T(1) + n^2 (2^k - 1) \quad 2n^3 - n^2$$

• Let prove by induction

$$\text{for } k=0, n=2^k=2^0=1 \quad T(1) = 2 \cdot 1^3 - 1^2 = 1 \quad \checkmark$$

$$\text{for } k=r, n=2^k=2^r \quad T(n=2^r) = 2 \cdot (2^r)^3 - (2^r)^2 \quad \text{hypothesis}$$

$$\text{for } k=r+1, n=2^{r+1} \quad T(n=2^{r+1}) = 2 \cdot (2^{r+1})^3 - (2^{r+1})^2 \quad \text{need to prove}$$

$$\begin{aligned} T(2^{r+1}) &= 8T\left(\frac{2^{r+1}}{2}\right) + (2^{r+1})^2 \\ &= 2^3 T(2^r) + (2^{r+1})^2 \\ &= 2^3 [2 \cdot (2^r)^3 - (2^r)^2] + (2^{r+1})^2 \\ &= 2^3 2 \cdot (2^r)^3 - 2^3 (2^r)^2 + (2^{r+1})^2 \\ &= 2(2^{r+1})^3 - 2 \cdot (2^{r+1})^2 + (2^{r+1})^2 \\ &= 2(2^{r+1})^3 - (2^{r+1})^2 \quad \checkmark \end{aligned}$$

$$T(n) = 2n^3 - n^2 = \Theta(n^3)$$

b) Recurrence relations using characteristic equation, find the time-complexity.

a) $T(n) = 5T(n-1) - 6T(n-2) + 4n - 3$ $T(0) = 1$ $T(1) = 2$

1) $T(n) - 5T(n-1) + 6T(n-2) = 4n - 3$

2) $p(r) = r^2 - 5r + 6 = (r-3)(r-2)$

3) $q(n) = 4n - 3 = 1^n (4n - 3)$

$q(r) = (r-1)^{1+1} = (r-1)^2$

4) $p(r)q(r) = (r-3)(r-2)(r-1)^2$ } characteristic equation
 $H(n) = A \cdot 3^n + B \cdot 2^n + C \cdot 1^n + D n 1^n$

5) Delete $p(r)$ part from $H(n)$

$R(n) = C + Dn$

6) Plug $R(n)$ into original equation

$C + Dn - 5[C + D(n-1)] + 6[C + D(n-2)] = 4n - 3$

$2C + 2Dn - 7D = 4n - 3$

$2Dn = 4n$ $D = 2$, $2C - 7D = 2C - 14 = -3$, $2C = 11$ $C = 11/2$

7) $H(n) = A \cdot 3^n + B \cdot 2^n + \frac{11}{2} + 2n$

8) $T(0) = 1 = A + B = \frac{11}{2} \rightarrow A + B = -\frac{9}{2}$

$T(1) = 2 = 3A + 2B + \frac{11}{2} + 2 \rightarrow 3A + 2B = -11/2$

} $A = \frac{7}{2}$
 $B = -8$

$T(n) = \frac{7}{2} 3^n - 8 \cdot 2^n + 2n + \frac{11}{2} = O(3^n)$

1) write equation like this (homogen part = non-homogen part)

2) write characteristic polynomial from homogen part $\rightarrow p(r)$

3) find equation from non-homogen part. $q(n) = 5^n s(n)$, $q(r) = (r-5)^{d+1}$

4) $q(r) \cdot p(r) =$ characteristic equation and the solution $\rightarrow H(n)$

5) Delete part of $H(n)$ coming from $p(r) \Rightarrow R(n)$

6) Plug $R(n)$ into original equation and solve constants

7) $R(n)$ is a particular solution

8) Use initial conditions with $H(n)$ find another constants.

$$5) T(n) = 5T(n-1) - 6T(n-2) + 2^n \quad T(0)=1, T(1)=2$$

$$1) T(n) - 5T(n-1) + 6T(n-2) = 2^n$$

$$2) p(r) = r^2 - 5r + 6 = (r-3)(r-2)$$

$$3) q(n) = 2^n \cdot 1^0$$

$$q(r) = (r-2)^{\infty} = (r-2)$$

$$4) p(r) \cdot q(r) = (r-3)(r-2)^2 \quad \text{characteristic equation}$$

$$H(n) = c_1 \cdot 3^n + c_2 \cdot 2^n + c_3 \cdot n \cdot 2^n$$

$$5) \text{Delete } p(r) \text{ part from } H(n)$$

$$R(n) = c_3 n \cdot 2^n$$

$$6) \text{Plug } R(n) \text{ to original equation and find } c_3 \text{ value}$$

$$c_3 n 2^n - 5[c_3 (n-1) 2^{n-1}] + 6[c_3 (n-2) 2^{n-2}] = 2^n$$

$$n 2^n \left[c_3 - \frac{5}{2} c_3 + \frac{6}{4} c_3 \right] + \frac{5}{2} c_3 - \frac{12}{4} c_3 (2^n) = 2^n$$

$$-\frac{2}{4} c_3 = 2^n \quad c_3 = -2$$

$$7) a \cdot 3^n + c_2 2^n - 2n 2^n = H(n)$$

$$8) T(0)=1 = c_1 + c_2$$

$$T(1)=2 = a \cdot 3 + c_2 \cdot 2 - 4 = (3a + 2c_2 = 6)$$

$$\left. \begin{array}{l} c_1 = 4 \\ c_2 = -3 \end{array} \right\}$$

$$T(n) = 4 \cdot 3^n - 3 \cdot 2^n - 2n 2^n = O(3^n)$$

$$c) T(n) = 4T(n-1) - 4T(n-2) \quad T(0)=1 \quad T(1)=2$$

$$1) T(n) - 4T(n-1) + 4T(n-2) = 0$$

$$2) p(r) = r^2 - 4r + 4 = (r-2)^2$$

$$4) (r-2)^2 \rightarrow H(n) = c_1 \cdot 2^n + c_2 \cdot n \cdot 2^n$$

$$8) T(0)=1 = c_1$$

$$T(1)=2 = c_1 \cdot 2 + c_2 \cdot 2 \quad \left. \begin{array}{l} c_1 = 1 \\ c_2 = 0 \end{array} \right\}$$

$$T(n) = 2^n = O(2^n)$$

7) Write recurrence relation following Algorithm 3, Algorithm 4 Analyze time complexity.

For Algorithm 3 $T(n) = T(n-2) + n + 1$ represents the exec time

$T(0) = 0, T(1) = 0 \rightarrow$ initial values

1) $T(n) - T(n-2) = n + 1$

2) $p(r) = r^2 - 1 = (r-1)(r+1)$

3) $q(n) = (n+1) 1^n$

$$q(r) = (r-1)^{1+1} = (r-1)^2$$

4) $p(r) \cdot q(r) = (r-1)^3 \cdot (r+1) \rightarrow$ characteristic equation

$$H(n) = c_1 \cdot 1^n + c_2 \cdot n \cdot 1^n + c_3 \cdot n^2 \cdot 1^n + c_4 \cdot (-1)^n$$

5) Delete $p(r)$ part from $H(n)$

$$R(n) = c_2 n + c_3 n^2$$

6) plug $R(n)$ into original equation

$$(c_2 n + c_3 n^2) - [c_2(n-2) + c_3(n-2)^2] = n + 1$$

$$n(c_2 - c_2 + 4c_3) + n^2(c_3 - c_3) + 2c_2 - 4c_3 = n + 1$$

$$c_3 = \frac{1}{4} \quad c_2 = 1$$

$$7) H(n) = c_1 \cdot 1^n + n \cdot 1^n + \frac{n^2 \cdot 1^n}{4} + c_4 \cdot (-1)^n$$

$$8) T(0) = 0 = c_1 + c_4 \rightarrow c_1 = -c_4 \quad c_1 = -\frac{5}{8}$$

$$T(1) = 0 = c_1 + 1 + \frac{1}{4} - c_4 \rightarrow 2c_4 = \frac{5}{4} \quad c_4 = \frac{5}{8}$$

$$T(n) = -\frac{5}{8} + n + \frac{n^2}{4} + \frac{5}{8}(-1)^n = O(n^2)$$

7)

For algorithm 4

We need to find recurrence relation

$$\left. \begin{array}{l} T(1) = 1 + T(0) = 2 \\ T(2) = 2 + T(1) = 4 \\ T(3) = 4 + T(2) = 8 \\ T(4) = 8 + T(3) = 16 \\ \vdots \\ T(n) = 2 * T(n-1) \end{array} \right\} T(n) = 2T(n-1), \quad T(0) = 1$$

Solution 1 (using characteristic equation)

$$\begin{array}{l} 1) T(n) - 2T(n-1) = 0 \\ 2) p(r) = r - 2 \\ 4) p(r) \cdot q(r) = r - 2 \rightarrow \text{characteristic equation} \\ H(n) = a \cdot 2^n \end{array}$$

$$8) T(0) = 1 = a \cdot 2^0 \rightarrow a = \frac{1}{2}$$

$$T(n) = \frac{1}{2} \cdot 2^n = 2^{n-1} = O(2^n)$$

Solution 2 (using substitution)

$$\begin{array}{l} T(n) = 2(2T(n-2)) \\ T(n) = 2(2(2T(n-3))) \\ \vdots \\ T(n) = \underbrace{2 \cdot 2 \cdot \dots \cdot 2}_{n \text{ terms}} \cdot \underbrace{T(0)}_{1} = 2^n \text{ (need to prove)} \end{array}$$

Prove by induction

$$\text{for } n=0 \quad T(0) = 2^0 = 1$$

$$\text{for } n=r \quad T(r) = 2^r \text{ (hypothesis)}$$

$$\text{for } n=r+1 \quad T(r+1) = 2^{r+1} \text{ (need to show)}$$

$$T(r+1) = 2T(r+1-1) = 2T(r) = 2 \cdot 2^r = 2^{r+1}$$

$$T(n) = O(2^n)$$

8) int findIndexofFirstone(A[], l, r)

A = 00000111 example

if $l \leq r$ {

$$m = \left\lfloor \frac{r+l}{2} \right\rfloor + 1$$

if $A[m] == 1$

return findIndexofFirstone(A, l, m-1)

else

return findIndexofFirstone(A, m+1, r)

} else {

return m-1 }

$$T(n) = T(n/2)$$

int cantores(A[], n)

return n - findIndexofFirstone(A, 1, n)

$$\begin{cases} T(n) = \log n + 1 \\ T(n) = \Theta(\log n) \end{cases}$$

Explanation of this algorithm time complexity.

$$T(n) = T(n/2) + 1$$

$$T(n) = T(n/4) + 1 + 1$$

$$T(n) = T(n/8) + 1 + 1 + 1$$

⋮

$$T(n) = \underbrace{T(1)}_{T(1)} + \underbrace{1+1+\dots+1}_{k \text{ terms}} = k+1 \Rightarrow T(n) = k+1 = \log n + 1 = \Theta(\log n)$$

$T(1) = 1$ (initial condition)

we assume that $n = 2^k$

(Best case, worst case, avg-case) all of them have a same time complexity.

That's why? Because this algorithm find first occurrence index for one. For this reason (unlike binary search), this function checks the whole range it has reduced.

Need to $\log n$ time for this operations.

what is the difference of binary search?

Binary search checks the key, it's equal to element will be terminated.

9) $int[]$ findTwoElements($A[], n$)

$p=0$ $q=n-1$

while $p < q$

if $A[p] + A[q] == x$

return $A[p], A[q]$

else if $A[p] + A[q] > x$

--q

else

++p

return

running time is $\Theta(n)$

Example

$p \downarrow$ $q \downarrow$

1	3	4	5	9
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 $A[]$ $n=5$ $x=8$
 $0 \ 1 \ 2 \ 3 \ 4$

step 1 ($A[0] + A[4] = 10$) $> (x=8)$

--q

step 2 ($A[0] + A[3] = 6$) $< (x=8)$

++p

step 3 ($A[1] + A[3] = 8$) $= (x=8)$

return $A[1], A[3]$

done!